



MARIO C. CALATAN

Utility Systems Engineer | GIS & SCADA Specialist | Smart Grid & Web GIS Developer

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PROFESSIONAL SUMMARY

Electrical Engineer with over 10 years of experience in electric distribution utilities, specializing in Smart Grid modernization, SCADA/GIS integration, Advanced Metering Infrastructure (AMI), and utility digital transformation initiatives. Experienced in leading cross-functional engineering teams, system modernization programs, and development of enterprise-grade utility applications that improve reliability, operational efficiency, compliance, and decision-making.

Strong hybrid background combining power systems engineering, utility operations, software development, databases, and AI-assisted technologies to deliver end-to-end solutions supporting future-ready electric utility operations.

CORE COMPETENCIES

Utility Engineering & Smart Grid

- Electric Distribution Planning & Design (up to 23kV)
- Smart Grid Technologies (SCADA, OMS, DMS, ADMS, AMI, FLISR)
- Distribution Automation and Utility Modernization
- Meter Data Management Systems (MDMS)
- Distribution System Reliability and Performance Analysis
- Utility Asset Planning and Infrastructure Development

Power Systems Engineering

- Load Flow Analysis
- Short Circuit Studies
- Arc Flash Analysis
- Protection Coordination Studies
- Feeder Capacity Assessment
- Fault Investigation and System Analysis
- Transformer Loading and Capacity Assessment
- Voltage Profile and Feeder Optimization

Relay & Protection Systems

- Protection relay coordination support
- Hands-on experience with ABB, GE, SEL, G&W, and Eaton equipment
- Recloser and feeder protection applications
- Fault isolation and switching strategies

Software Development & Automation

- Web-based Utility Systems Development
- Full-stack Development (PHP, Python, JavaScript, React, Node.js)
- Utility Data Integration and Analytics
- Database Design and Optimization
- AI-Assisted Development and Automation

Databases

- PostgreSQL
- MySQL/MariaDB

- Oracle
- Microsoft SQL Server

Communication Protocols

- DNP3
- Modbus
- SCADA Data Communications

AI & Emerging Technologies

- GitHub Copilot
 - Claude AI
 - GPT/OpenAI
 - Base44 AI-assisted development tools
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SOFTWARE & TECHNICAL TOOLS

Smart Grid Platforms

- Survalent
- QEI
- SEL Technologies

Planning & Engineering Tools

- Synergi Electric
- ETAP
- PVSyst

GIS & Design Tools

- QGIS
- PostGIS
- ArcGIS
- AutoCAD
- SketchUp
- Canva

Productivity Platforms

- Microsoft Office Suite
(Word, Excel, PowerPoint)
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KEY SYSTEMS & PLATFORMS DEVELOPED

Led the design and development of multiple in-house enterprise applications supporting utility operations and digital transformation initiatives:

- Network Management System (NMS)
- Web GIS Platform with modules including:
 - Vegetation Management
 - Damage Assessment
 - Thermal Scan Inspection Management
- Meter Installation Management System
- Meter Replacement Management System
- Technical Repository System
- Compliance Reporting System
- Transformer Management System

- Meter Data Management System (MDMS)
 - Calibration Management System
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PROFESSIONAL EXPERIENCE

Smartgrid Engineer / System Integrator / Web Developer

BENECO | 2017 – Present

Overhauled SCADA systems for OMS/DMS readiness and modernized the utility GIS platform. AMI development such as Meter Data Management System. Connecting Kwhr Meters through API and integrating it to SCADA through Modbus/TCP protocol. Operational Technology - Integration of Circuit Breakers, recloser, Load Break Switch, Disconnect Switches, Protection Relays through DNP3 and Modbus

- Integrated GIS, OMS, and SCADA data for the company's partial ADMS project.
- Participated in pilot projects for over three different AMI brands to enhance Advanced Metering Infrastructure.
- Developed an Automatic Meter Reading (AMR) system tailored for large load customers.
- Successfully integrated AMR meters with SCADA using Modbus TCP for improved data management.

Designed and developed multiple web-based enterprise and workflow systems that simplify, standardize, and automate utility business processes and engineering operations. Projects include a Network Management System (NMS), Web GIS Platform, Meter Installation and Replacement Management Systems, Meter Data Management System (MDMS), Transformer Management System, Compliance Reporting System, Technical Repository System, and Calibration Management System. Utilized PHP, Python, JavaScript, React, Node.js, PostgreSQL, MySQL, and Oracle to develop scalable, data-driven applications. Focused on creating workflow-driven solutions that improve process consistency, operational efficiency, collaboration, reporting, and decision-making while supporting utility digital transformation and smart grid modernization initiatives.

System Planning & Design Officer

BENECO | 2024 – Present

- Lead planning and design activities for electric distribution network development and infrastructure expansion initiatives
- Manage engineering teams responsible for CAPEX/OPEX planning, inspections, forecasting, and project prioritization
- Conduct power system studies including:
 - Load Flow Analysis
 - Short Circuit Studies
 - Protection Coordination Studies
 - Reliability Analysis
- Lead As-Planned and As-Built engineering activities
- Develop technical reports and management presentations supporting strategic decisions
- Serve as Chairman of the Technical Working Group under the Bids and Awards Committee (BAC) responsible for:
 - Technical specification review
 - Bid evaluation
 - Recommendation preparation

- Support long-term infrastructure planning and utility modernization initiatives
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Special Equipment & Metering Officer

BENECO | 2020 – 2024

- Led implementation initiatives for Advanced Metering Infrastructure (AMI) including Proof-of-Concept, pilot, and deployment phases
 - Developed and implemented multiple in-house utility systems supporting metering and operational functions
 - Led deployment of Meter Data Management System (MDMS) solutions
 - Performed testing and technical evaluation of:
 - Meters
 - Transformers
 - CT/PT systems
 - Line equipment
 - Supported testing and commissioning activities
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Construction & Maintenance Officer

BENECO | 2019 – 2020

- Managed 24/7 field operations and maintenance activities
 - Led emergency response coordination and service restoration efforts
 - Supervised field crews and coordinated operational priorities
 - Provided technical guidance for construction and maintenance activities
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SCADA & GIS Development Engineer

BENECO | 2013 – 2019

- Modernized SCADA infrastructure supporting OMS/DMS readiness initiatives
 - Led GIS enhancement projects and utility network modeling improvements
 - Integrated operational systems to improve situational awareness and data visibility
 - Supported Smart Grid and utility digital transformation initiatives
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GIS Data Engineer

BENECO | 2011 – 2013

- Conducted GIS data gathering and validation activities
 - Supported GIS implementation and utility asset integration projects
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ADMINISTRATIVE & LEADERSHIP EXPERIENCE

- CAPEX and OPEX preparation, justification, and monitoring
- Team leadership and workforce planning
- Stakeholder coordination across operations, vendors, regulators, and communities
- Procurement support and resource planning
- Technical report preparation and executive presentations
- Project documentation and compliance reporting
- Policy implementation and ISO/IMS process alignment

- BAC Technical Working Group leadership
 - Mentoring and cross-functional coordination
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EDUCATION

Master in Urban Management (*Ongoing*)
Benguet State University | 2020 – Present

Bachelor of Science in Electrical Engineering
Saint Louis University | 2006 – 2011

ADDITIONAL INFORMATION

- Willing to relocate internationally with family subject to employer-supported arrangements
- Strong background in Smart Grid and utility digital transformation initiatives
- Combines engineering leadership with software and analytics expertise
- Open to utility engineering, planning, smart grid, and digital transformation opportunities